

Fernando J. Soto

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Technical Skills

- Languages: C#, Python, Java, JavaScript, PHP
- Backend & APIs: ASP.NET Core, Flask, RESTful APIs
- Frontend: React, HTML, CSS
- Databases: SQL, MongoDB, Redis
- Testing: Cypress, Selenium
- Tools: Git, Bash

Work Experience

Software Engineer (Remote, US)

Paylocity | August 2024–April 2026 | 3,000 req/s • 7.5x performance • 8x query reduction

- Achieved 7.5x performance gains and 8x query reduction in high-traffic Time Entry APIs by eliminating redundant permission checks and reworking inefficient SQL query patterns.
- Designed and enhanced bulk Time Entry and Pay Type APIs (V2/V3), handling complex modifiers, validation-heavy edge cases, and versioned endpoint behavior.
- Prevented a regression impacting millions of users by proactively querying production data for misconfigured accounts before a settings change shipped, surfacing the risk during refinement instead of after release.
- Diagnosed and fixed a defect causing audit trail entries to silently fail on POST/PATCH across three Time Entry API versions, restoring compliance-critical data integrity.
- Migrated WebTime and TimeApi off a shared build pool onto dedicated infrastructure, reducing contention and improving build reliability for critical services.

Associate Software Engineer (Remote, US)

Paylocity | July 2022–August 2024 | 22.5% of user base covered

- Built automated test coverage for a product suite used by ~22.5% of the user base, improving release stability.
- Migrated testing from Selenium to Cypress and established unit testing patterns adopted as team standard, reducing flakiness and improving development speed.
- Identified and fixed a defect impacting 18% of users by tracing a marketplace-side query pattern against TimeAPI, restoring functionality without requiring downstream changes.
- Improved API documentation standards, including formalizing response codes, payload shapes, and validation-ordering behavior for high-traffic bulk endpoints.

Software Engineer Intern (Remote, US)

Paylocity | May 2021–August 2021 | 15% page load improvement

- Improved page load performance by 15% by diagnosing and resolving critical frontend/backend issues; contributed to analytics tracking and database modernization efforts.

Education

Bachelor of Science in Computer Science and Mathematics

Dominican University—May 2022

Projects

Ultra Cycling Planner | TypeScript, React, FastAPI, Python, Docker

- Engineered a GPX processing pipeline that decimates tracks of 100k+ points to a configurable render resolution, then dynamically re-samples at higher fidelity on zoom—avoiding aliasing artifacts while keeping the DOM lean for client-side rendering.
- Computed per-split terrain metrics (elevation gain/loss, grade, steep-distance %) directly from raw GPS coordinates, powering race strategies that contributed to a 2nd-place finish at the Mishigami Challenge.
- Integrated third-party mapping/location APIs (OpenStreetMap, Google Maps/Places) with configurable caching, request debouncing, and provider failover to respect rate limits and improve reliability.